

SYED OSAMA HUSSAINI

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PROFESSIONAL SUMMARY

Mechanical Engineering graduate with hands-on experience in **HVAC, manufacturing, and preventive maintenance** across **oil & gas, shipyard and building services** industries. Proficient in **SolidWorks, ANSYS, CAD/FEA/CFD and Carrier HAP**; seeking to drive innovation in **energy-efficient home appliance engineering** at **PEL**.

EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI) *Topi, Pakistan*
B.S. Mechanical Engineering — CGPA: 3.63/4.00 *2022 – 2026*

- Awarded **5 Dean's Honor Roll Certificates** (Semesters 1, 2, 5, 6 & 7) for academic excellence.

Whales College *Karachi, Pakistan*
*A Levels: Mathematics, Physics, Chemistry — 3 A*s* *2020 – 2022*

WORK EXPERIENCE

Karachi Shipyard & Engineering Works Ltd. *Karachi, Pakistan*
Mechanical Engineering Intern *Jul 2025 – Aug 2025*

- Assisted in **valve overhauling, hydraulic systems inspection, CNC machining**, and mechanical quality checks.
- Contributed to **preventive maintenance planning**, improving equipment reliability for rotating machinery.
- Gained practical exposure to **manufacturing processes** and shipyard engineering workflows.

Mari Energies – MSD RIG-04 *Daharki, Pakistan*
Mechanical Engineering Intern, Drilling Rig Operations *Jun 2025 – Jul 2025*

- Assisted in maintenance and inspection of **Drawworks, Top Drive, Mud Systems**, and BOP components.
- Applied **HSE frameworks** including PTW, LOTO, and Well Control Awareness protocols.

Paragon Constructors Ltd. – Emaar North Tower Project *Karachi, Pakistan*
Mechanical Intern, Building Services Systems *Jul 2024 – Aug 2024*

- Worked on **HVAC systems, pumping systems**, and mechanical utilities in a high-rise construction project.
- Assisted with **QA/QC documentation, ducting, insulation, VRF systems** and site inspections.

PROJECTS

[FYP] FLEX-SENS: Flexible Nanoparticle-Infused Tactile Sensor for Smart Systems *2025 – 2026*
1st Position – Mechanical Engineering Department, GIKI

- Developed a **flexible capacitive tactile sensing system** using **silver conductive ink electrodes** and **RTV silicone-ZnO nanocomposite** dielectric for smart sensing applications.
- Integrated **NE555-based signal acquisition** circuit and applied **machine learning-based load prediction** algorithms.

HVAC Design and Analysis of an 8-Story Commercial Building *2024*
Thermal Systems Design Project

- Designed a complete **HVAC system** using **Carrier HAP** software per **ASHRAE standards**; performed **cooling load calculations, duct sizing**, and energy optimization.
- Delivered **energy-efficient HVAC solutions** ensuring thermal comfort and system performance compliance.

Design and Exergy Analysis of a 10 kW Organic Rankine Cycle (ORC) *2024*
Thermodynamics and Energy Systems Project

- Designed a **waste heat recovery ORC system**; conducted **thermodynamic/exergy analysis** to maximize efficiency.
- Identified key irreversibility losses and proposed **energy optimization strategies** for sustainable power generation.

LEADERSHIP EXPERIENCE

Head of Documentation Sub-Team – Team Foxtrot, GIKI *2023 – 2024*
Autonomous UAV Development Team

- Led **technical documentation and workflow coordination** for autonomous UAV design and development projects.
- Supervised team members and maintained **engineering reports, technical records**, and evaluation documentation.

TECHNICAL SKILLS

CAD/Simulation: SolidWorks, AutoCAD, ANSYS, COMSOL, CATIA V5, CAD Modeling, FEA, CFD

HVAC & Thermal: Carrier HAP, HVAC Systems Design, Thermodynamics, Heat Transfer, Cooling Load Analysis

Manufacturing & Maintenance: Manufacturing Processes, CNC Machining, Hydraulic Systems, Pump Systems

Engineering & Analysis: Product Design, Mechanical Design, Exergy Analysis, Sensor Development, Flexible Electronics

CERTIFICATIONS

IWCF Level 1 – Well Control Awareness | **CATIA V5** – Beginner to Advanced Level | **IHRDC** – Valves: Electric and Hydraulic Actuators | **IHRDC** – Centrifugal Pump Basics and Troubleshooting